Session Summary: 44.1 and 48 kHz Formats

Goal and Result

The ESP32-S3 + PCM5102 card only supported 44.1 kHz / 16 bit. Now Windows 11 allows selecting either 44.1 or 48 kHz (16 bit stereo), and the audio path switches on the fly. Switching works in both directions.

Technical Findings

- The ceiling is 48 kHz. The USBAudioCard class in core 3.3.12 rejects higher rates on the UAC1 path (USB Full Speed), and TinyUSB buffers are built into precompiled libraries. 88.2 and 96 kHz would require rebuilding the core or a different source than USB UAC1.
- Core 3.3.12 has only a single frequency in the descriptor. The macros in USBAudioCardDescriptors.h already accepted a list, so the fix in USBAudioCard.cpp was small.

Code Changes

- USBAudioCard.cpp (copy from the core): USB_UAC1_RATES list (44100, 48000) in the descriptor, rejection of frequencies outside the list, check of the starting frequency.
- usb_soundcard.cpp: handling of the frequency change event, which does not count toward the volume/mute counter.
- audio_source.h, audio.cpp: new optional get_sample_rate pointer polled in AUDIO_Poll(). The actual change is performed by the output thread in audioFill: it retunes the I2S clock, flushes the FIFO, sets a new drift target, and does a re-prefill.
- audio_out_i2s_esp32.cpp: AUDIO_OUT_SetSampleRate().
- fifo.*, drift.*, audio_spectrum.*: FIFO_Flush, DRIFT_SetTarget, SPEC_SetRate.
- audio_config.h: frequency list and maximum. The target FIFO level is now in milliseconds (46 ms), and the buffer has grown to 32 kB.
- New sample_rate field in AUDIO_Stats_t and AUDIO_GetSampleRate() function.

Confirmed on Hardware

- fs in STAT matches the selection in Windows.
- Blocks per second: 172.4 at 44.1 kHz and approx. 188 at 48 kHz, i.e. the I2S clock switches.
- Drift correction approx. 0 ppm, OV 0 UF 0, FIFO stable.
- UF 1 after switching and a slight click is the expected result of the momentary stream interruption by Windows. You accept this.

Installation and Best Practices

- Keep core files (USBAudioCard.cpp) substituted in the core, not in the sketch, to avoid multiple definition errors.
- After a core update, compare the copy with the new version from the core.
- The frequency list must be the same in USB_UAC1_RATES and AUDIO_SAMPLE_RATE_LIST.

Explained Along the Way

- Spectrum analyzer at 48 kHz. Bands and decay time are recalculated. CPU load increases by approx. 9% because there are more frames, while the time of a single FFT (4416 µs) does not change. That is

why fft_us does not distinguish frequencies.

• I advised against undersampling the spectrum buffer: the gain is about 9%, but the cost is artifacts in the upper bars. The time hop remains unchanged, as per your decision.

To Do Later

• Longer 48 kHz test (a few minutes or more): watch FIFO min/max, UF and OV. For 44.1 kHz you have already verified this over hours.

• Previously deferred: sporadic watchdog resets (IDLE0, usbd task), behavior after prolonged stream inactivity, and optional conversion of the spectrum hop to time-based.